

Standard Operating Procedure

SCADA Parameter Excursion Response — Fermentation & Utility Systems

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Department	Manufacturing / QA	Classification	GxP Critical
Author	Process Engineering	Approved By	Head of Quality

1. Purpose

This procedure defines the corrective response when a monitored SCADA parameter departs its validated control band during the fermentation (growth) phase or associated utility operation. It applies to the five critical parameters supervised by the BioSCADA AI system: reactor temperature, pH, differential pressure, water-for-injection (WFI) conductivity, and cleanroom relative humidity.

2. Scope

Applies to all fermentation batches of Omez, Nise and Mintop and to the supporting WFI and HVAC utilities. Alarm-zone excursions are handled by the responsible operator or automated agent. Trip-zone excursions require Quality Assurance notification before any corrective action is executed.

3. Definitions

Term	Definition
Control band	The validated operating range; normal operation, no action required.
Alarm zone	Outside control but within safety limits; corrective action permitted.
Trip zone	Outside safety limits; QA e-signature obligation before action.
Residence time	Approximate process time constant used to pace adjustments (~90 s).
e-Signature	21 CFR Part 11 compliant electronic signature with signer identity and meaning.

4. Control Band Reference

Parameter	Asset	Control Band	Alarm Band	Trip Band
Reactor Temperature	BR-12	36.5 – 37.5 °C	36.0 – 38.0 °C	35.5 – 38.5 °C
pH	BR-12	6.8 – 7.2	6.6 – 7.4	6.4 – 7.6
Differential Pressure	FIL-07	100 – 110 kPa	96 – 114 kPa	92 – 118 kPa
Conductivity (WFI)	WFI-02	700 – 900 µS/cm	640 – 960 µS/cm	600 – 1000 µS/cm
Humidity (Cleanroom)	CR-A1	40 – 55 %	36 – 59 %	32 – 63 %

5. Procedure — Reactor Temperature (upward excursion)

Step 1. Confirm the reading against redundant probe TT-1202B. If the two probes disagree by more than 0.3 °C, treat as a probe fault and raise a deviation rather than adjusting the process.

Step 2. Reduce the coolant-loop setpoint in increments of no more than 4 percent. Wait one full residence time (approximately 90 seconds) between increments and observe the trend. Over-correction produces an undershoot that is harder to recover than the original excursion.

Step 3. Verify chiller CW-3 discharge temperature is at or below 8 °C. If discharge temperature is elevated, the chiller is the limiting factor and further coolant reduction will not help.

Step 4. Read heat-exchanger differential pressure. If it exceeds 35 kPa, fouling is indicated; raise a maintenance work order for heat-exchanger cleaning and hold the batch under enhanced monitoring.

Step 5. If temperature has not returned to the control band within 15 minutes of the first adjustment, escalate to the Production Lead and initiate a time-out-of-spec deviation record.

6. Procedure — pH (GxP-critical, QA e-signature mandatory)

Critical constraint: All pH setpoint changes require QA electronic signature before execution under 21 CFR Part 11. No autonomous pH setpoint adjustment is permitted by any automated system. An agent may prepare and recommend the adjustment but must hold execution pending signature.

Step 1. Confirm probe calibration age. If last calibrated more than 14 days ago, or the millivolt offset exceeds 12 mV, treat the reading as suspect and verify with an offline sample before dosing.

Step 2. Stage base dosing pump P-2 at an increment of no more than 3 percent of nominal dose rate. CO₂ accumulation is the most frequent driver of downward pH drift in the growth phase.

Step 3. Confirm agitator speed is at setpoint. Poor mixing produces local pH gradients the probe reads as a bulk excursion.

Step 4. Await QA electronic signature before committing the dose-rate change.

Step 5. Observe for two residence times after the change. Do not stack a second adjustment before the first has fully expressed.

7. Procedure — Differential Pressure (rising)

Step 1. Initiate a filter differential-pressure integrity check on FIL-07. Rising dP most commonly indicates progressive filter blockage.

Step 2. Open exhaust vent V-12 in increments of up to 10 percent to relieve headspace pressure while filter condition is assessed.

Step 3. Compare current dP against the bubble-point threshold recorded at installation. If the filter fails bubble point, change the filter and raise a work order before continuing.

Step 4. Check gas exhaust flow. A restricted exhaust presents identically to a blocked filter at the transmitter but requires a different remedy.

8. Procedure — WFI Conductivity

Step 1. Cross-check WFI feed composition against the approved batch recipe. Feed composition error is the leading cause of conductivity excursion and is correctable at source.

Step 2. Read total organic carbon. If TOC correlates with the conductivity rise, trigger loop hot-recirculation per the sanitisation method.

Step 3. Review the resin-bed regeneration cycle count. If it exceeds the validated limit of 150, the resin bed is exhausted and regeneration must be scheduled before the loop returns to service.

Step 4. Conductivity outside 600–1000 µS/cm renders the water non-conforming for injection use. Isolate the loop and quarantine affected material.

9. Procedure — Cleanroom Humidity

- Step 1.** Raise or lower the HVAC fan-speed setpoint in increments of no more than 6 percent. Fan speed is the fastest-acting control on cleanroom RH.
- Step 2.** Inspect cooling-coil temperature. A coil running warm cannot condense sufficient moisture and will not hold the RH band regardless of fan speed.
- Step 3.** Read HEPA differential pressure. A loaded HEPA restricts airflow and presents as an RH excursion.
- Step 4.** Review the door-open count for the preceding hour. Repeated openings produce transient RH spikes that are not equipment faults and should not trigger a maintenance action.

10. Revision History

Rev	Date	Description	Approved
04	02-Jul-2023	Added WFI conductivity procedure	H. Quality
05	18-Nov-2023	Humidity door-count clarification	H. Quality
06	11-Mar-2024	pH e-signature constraint made explicit	H. Quality